



Hydrodomus

Decentralizing the Hydrogen Grid

"EV wall charger for hydrogen"



LA
FRENCH TECH
GRAND PARIS



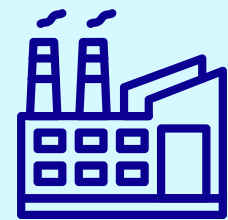
The problem



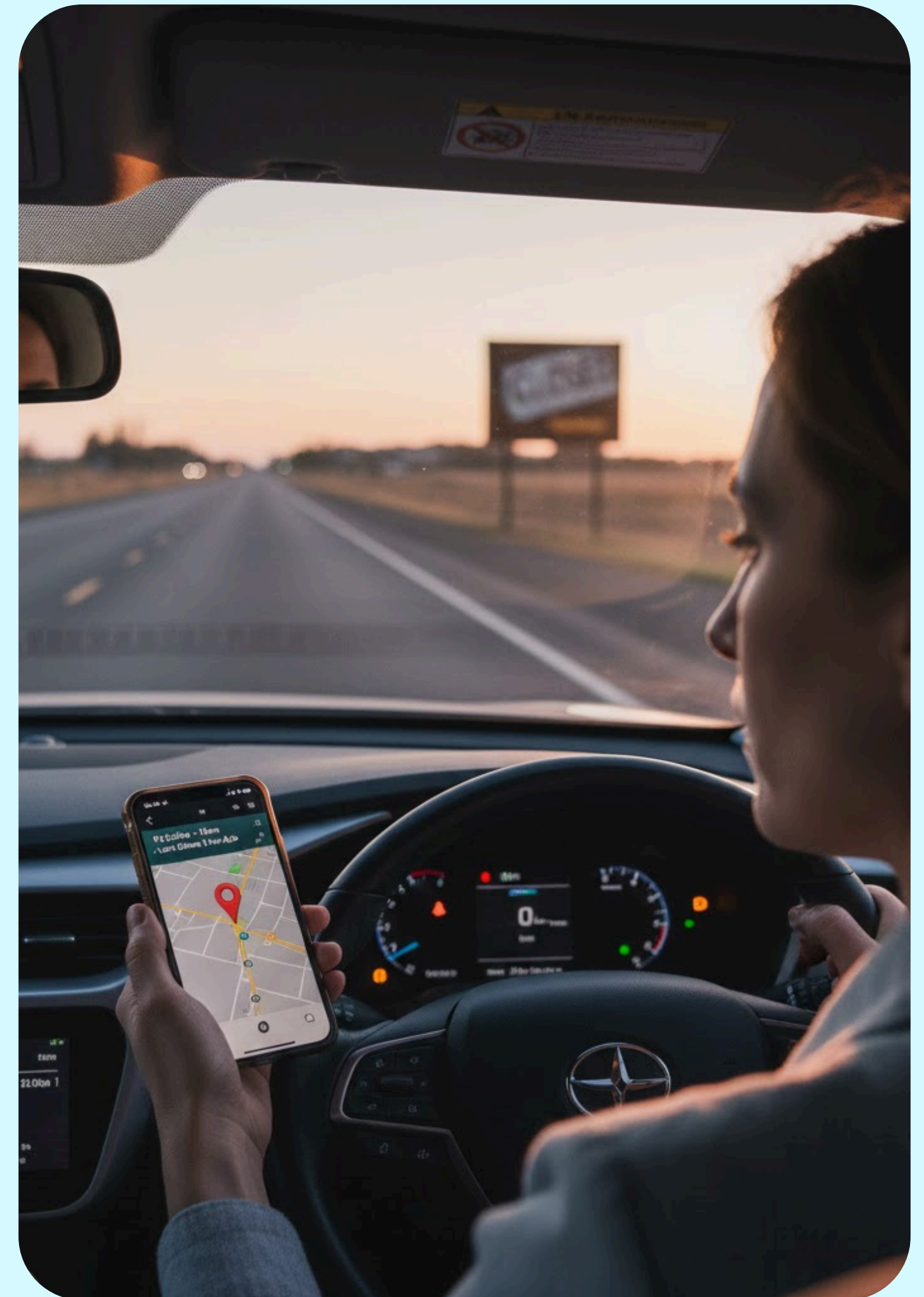
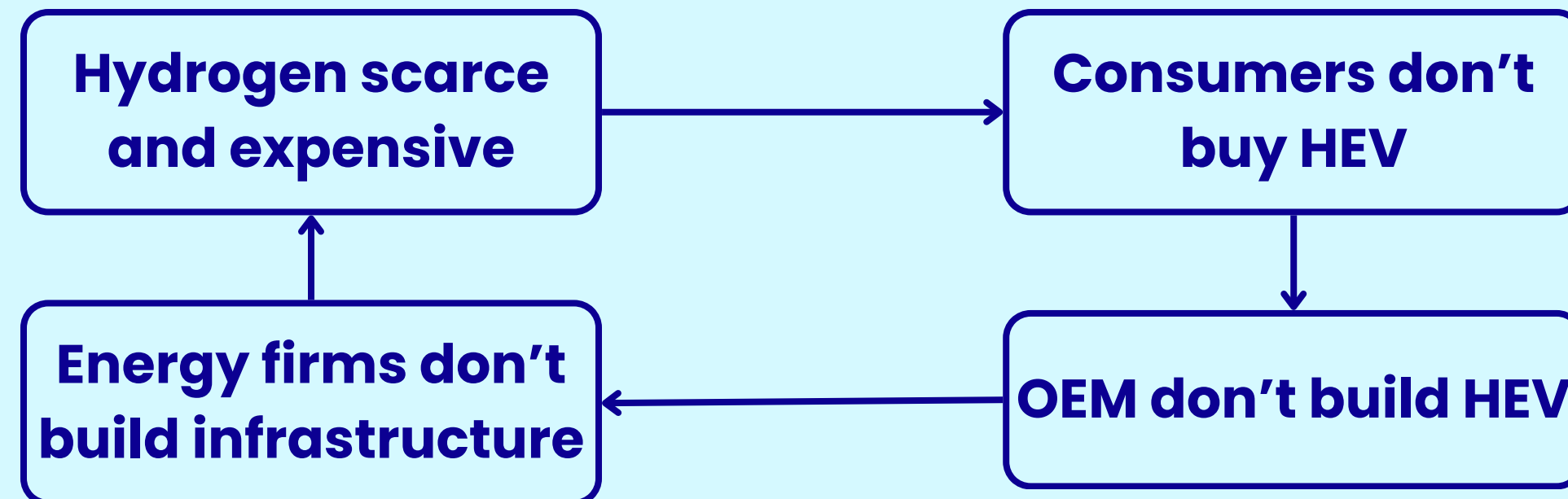
Poor **accessibility** (few stations) and high **cost**



85% of hydrogen is not **green** in the stations



Dependent on **external supply** or **big investments**



The solution



Hydrodomus concept :
"EV wall charger for hydrogen"



Only uses water and electricity:
no green-house effect gases



The end-user is fully autonomous:
they own the supply.



High safety of the system,
following every regulations



The Tech: Efficient, Safe & Scalable

Numbers :

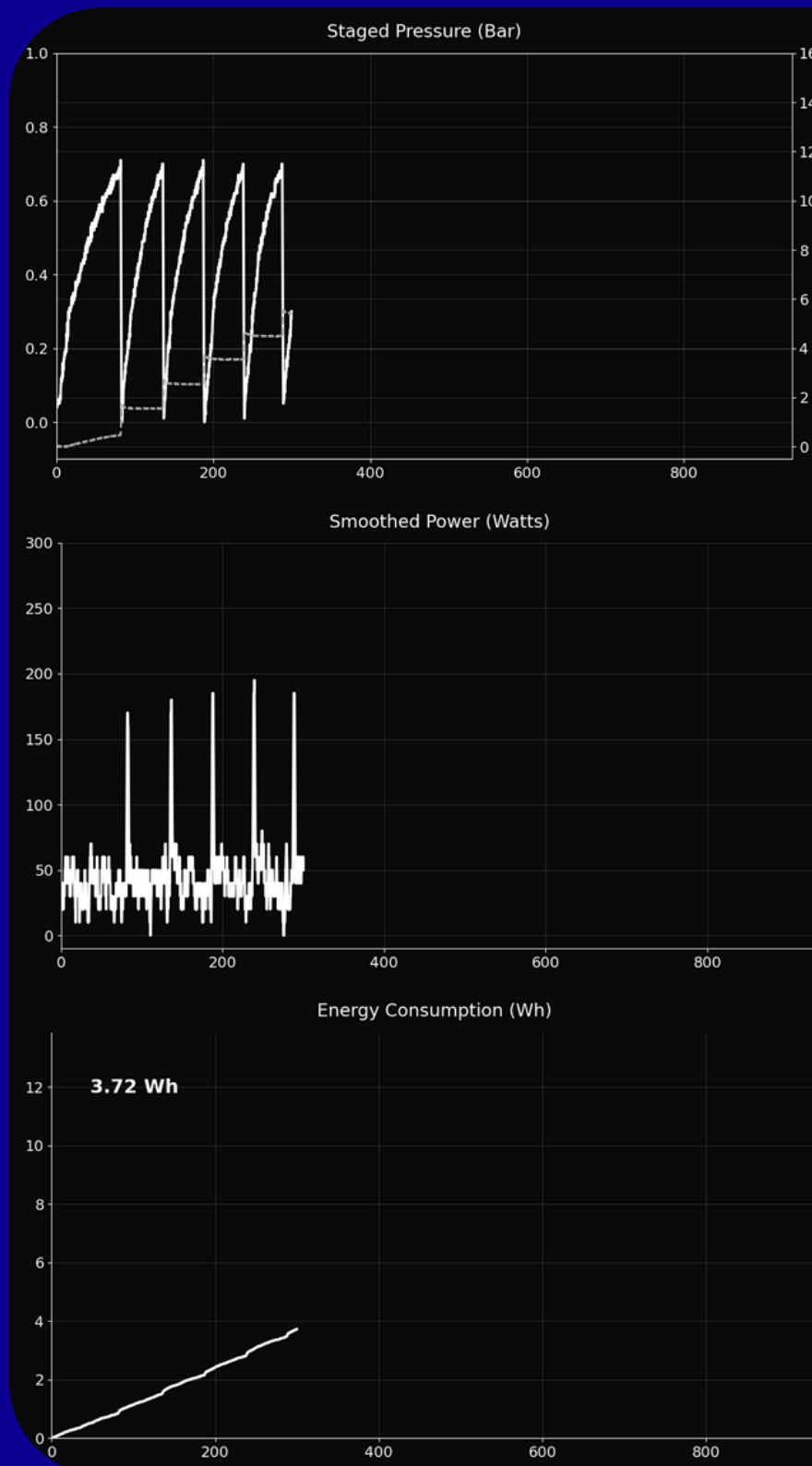
Efficiency: ~65%

Pressure: 700 bar

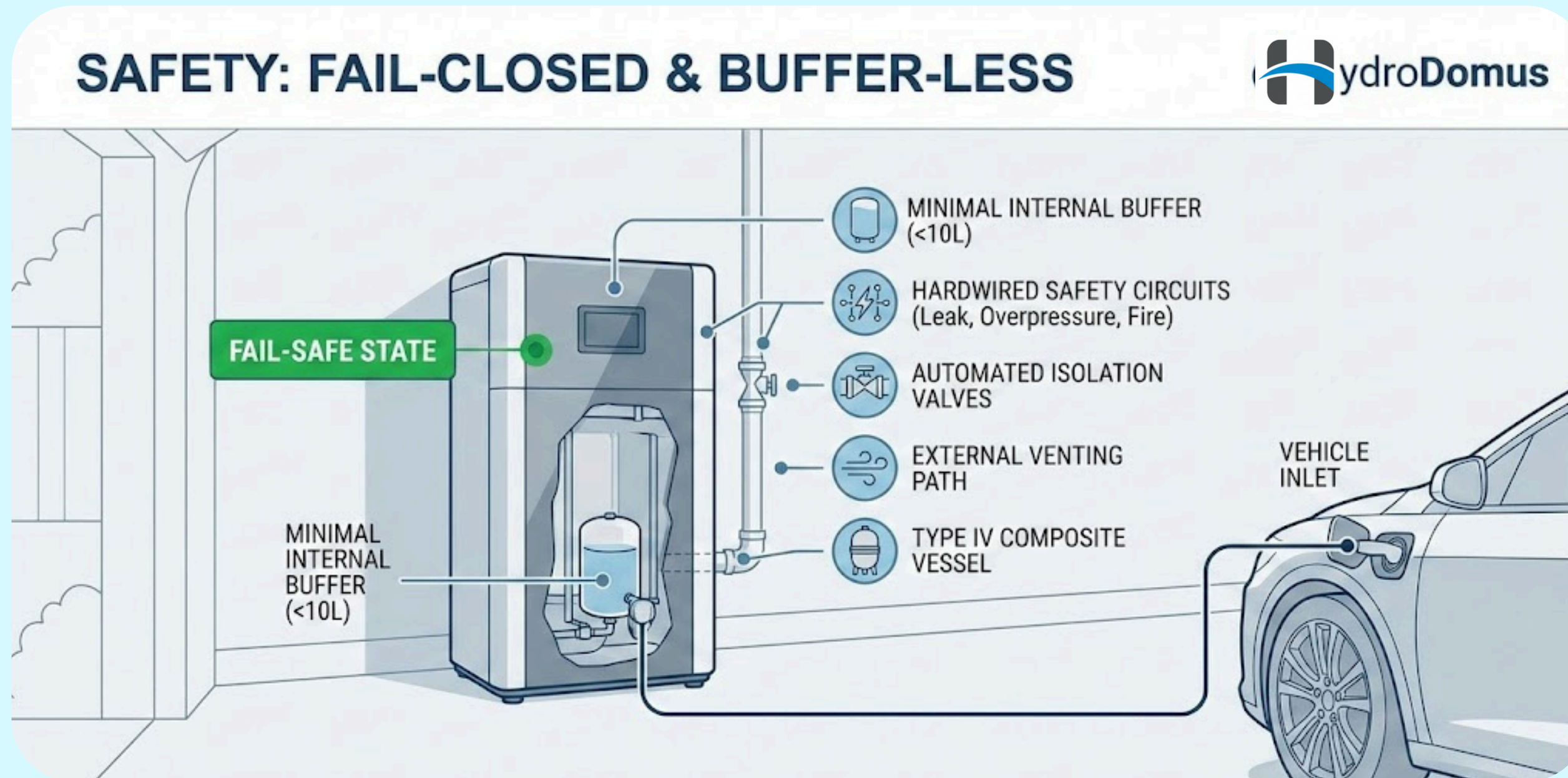
Purity: 99,99%

Refueling time ~3 minutes

1kg of H₂ generated every 24h



Safety



Most of the norms and certifications will be met by our suppliers

Norms and certifications :

- Hydrogen fueling : **ISO 19880**
- Refueling interface : **ISO 17268 / SAE J2600 / SAE J2601**
- Explosive atmosphere : **ATEX 2014/34/EU**
- PEM : **ISO 22734**
- Storage : **Type III / IV**
- Pressure Equipment : **PED 2014/68/EU**
- Machinery : **2006/42/EC**
- Electricity : **2014/30/EU**

The prototype

- **Status:** Subscale bench prototype built and operational.
- **Tech Validated:** PEM generation (6 L/h), multi-stage filtration, and 2-stage mechanical compression (14 bar).
- **Strategy:** Subsystems decoupled and tested independently to ensure maximum safety and low-cost validation before full 700-bar integration.



Market validation

Hydrogen is the ecological solution to transport but cars don't sell because there is not many pumps available.

Production is ramping up and the competition in hydrogen generation remains low in that field.

Public H2 price : 15–20€/kg

Home H2 production : 4–7€/kg

Context	Our opportunity
Big car companies don't sell their Hydrogen cars	Because there is no infrastructure to refuel
Big brands are scaling production	The market need for hydrogen refuel will increase
Cost of Hydrogen rises	We offer a much more affordable refuel
BMW & Toyota building Gen3 fuel cells	Shift from "Early Adopters" to "Series Production"

Market Size

As of today less than 100k Hydrogen cars have been sold.

Expected much greater numbers soon.

Huge market in the automotive but also to consider trucking, constructions, farming, etc...

The Hydrogen revolution is very similar to the EV boom, it grew **exponentially**.

Metric	Value	Reference
TAM	€150B+	Global Hydrogen Infrastructure & Home Energy Storage (2040 Horizon)
SAM	€15B	2.5M Passenger FCEVs projected by 2030 (Light Duty Vehicles).
SOM	€250M	High-utilization fleet/taxi drivers in EU/US (Immediate 3-year target).

Unit Economics

Manufacturing Cost — Unit #1	20,000€
Manufacturing Cost — Unit #1,000	8,000 €
Target Retail Price	11,500 €
Gross Margin	30 %
Fully loaded cost / kg H ₂ (consumer)	5 €/kg
Public station price / kg H ₂	15–20 €/kg
Customer payback period	100 refills
Long Term Target Retail Price	5,000 €

“EV wall charger for H2”

Domestic Hydrogen generator :

- Connect it to electricity and water supply.
- Plug the nozzle to your car.
- It generates Hydrogen and sends it directly in the car bottle.
- Unplug when you are ready.
- Safety comes before efficiency.



How it works



1. Generate Hydrogen gas by electrolysis
2. Compress the gas at 700 bars
3. Send it in the bottle inside the car



Business Model

OEM Bundling : Partner with car manufacturers to sell it as an option coming with the car

The customer leaves the dealership with a car and a refueling station.

One-time installation fee + recurring yearly sanity check and maintenance.

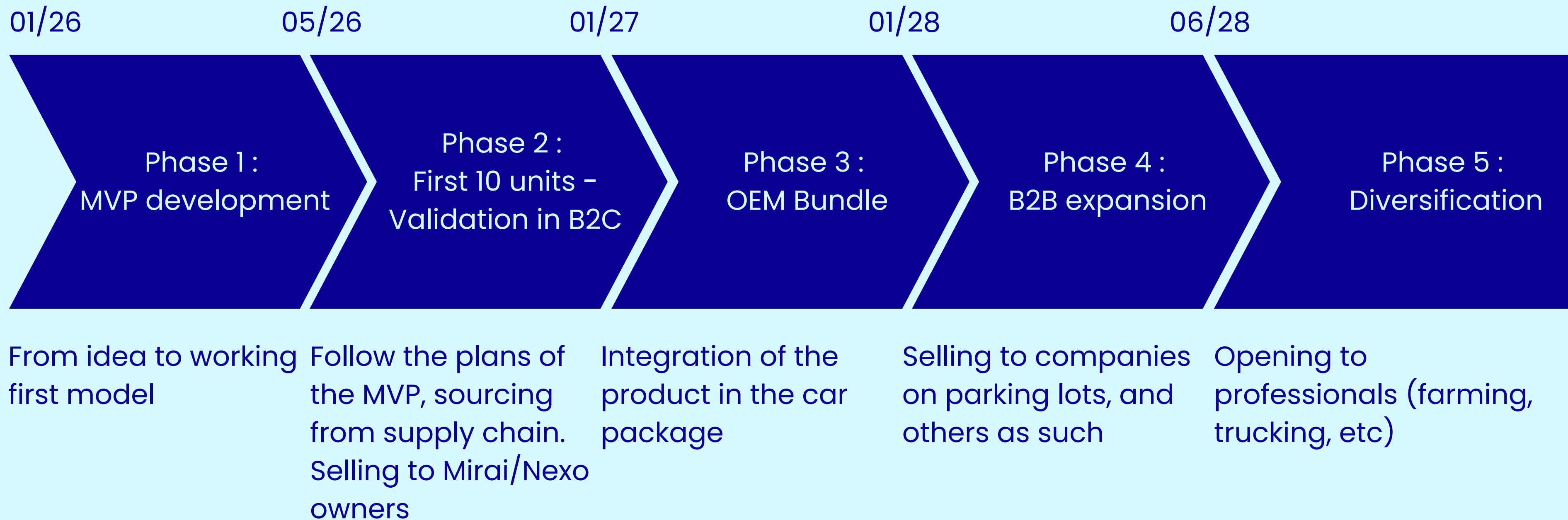
Remote monitoring and Software Updates.



Benchmark

	Public Stations (Air Liquide/HRS)	Toyota Cartridges (Prototype)	HydroDomus	Battery EV
Convenience	Low (Must travel to station)	Medium (Manual handling/swap)	High (Refuel while you sleep)	Medium (overnight charge, grid-dependent)
Infrastructure Cost	High (~€3M+ per station)	High (Logistics & swap hubs)	Low (Uses existing grid/water)	Low for light use; grid upgrades needed for heavy-duty
Fuel Source	Often Grey/Industrial H ₂	Varies (logistics-dependent)	100% Green (Self-generated)	Grid mix (30–50% fossil in EU avg.)
User Experience	"Gas Station" model	"Propane Tank" model	"EV Home Charging" model	Simple UI; limited for heavy-duty or off-grid
Logistics	High-Intensity Industrial	Complex Circular Logistics	Zero Logistics. Decentralized Production	Grid infrastructure upgrade required for heavy-duty

Go-To Market Plan



The Founding team



Yannick Durindel
Business development
& Engineering

MSc Hydrogen engine,
ICAM graduate,
Ex developer @ Eyalights
Consultant at mc2i

Fundraising

Use of Fund	
Hardware / 100 Pilot Units	1,0 m€
Safety Certification (ATEX, ISO, PED)	400 K€
Team Hires (Engineering + Sales)	1,4 m€
Operations & Legal	200 K€
Total	3 m€

The Ask	
Raising	0 €
Round	Seed
Runway	12 Months
Current Status	MVP in development
Lead Investor	N/A
Closing	3 Million €